INDEPENDENT RESEARCH PROTOTYPE · INSPIRED BY THE CESI LINEACT / IRIT 2027 INTERNSHIP TOPIC

HybridMobility-Coop

Who takes the car, and who takes the child to school? A small, reproducible benchmark of household agents that share scarce resources and face unexpected morning events. It compares rule-based, LLM-based and hybrid decision-making, with and without negotiation between the agents.

`Python, standard library only` `local LLMs via Ollama (qwen2.5:3b, qwen3:4b)` `120 paired test situations` `17 automated tests` `synthetic data`

MAIN RESULT

The best design used the LLM **only to read the unexpected news** and turn it into a fact. Simple rules and a negotiation protocol then made the decisions. It came closest to a perfect plan and beat a central coordinator, while calling the LLM far less than agents that let the LLM decide everything.

6.6
regret of the best design
(central coordinator: 31.3; LLM decides: 87.1)

0.6
LLM calls per morning
(LLM decides everything: 3.9)

74/80
news sentences read correctly
on a held-out test, 20/20 traps ignored

The questions

The research line this builds on (Vu, Gaudou & Oberoi 2025; Bru, Oberoi & Gaudou 2026) simulates how people in Toulouse choose to travel, using LLM-based "generative agents". The 2027 internship asks for two extensions, and this prototype tests both on a small, fully traceable problem:

1 · Hybrid decisions

Calling an LLM for every decision of thousands of agents is expensive. Can rules handle familiar situations, with the LLM used only when something unexpected happens? What does each approach gain and cost?

2 · Agents that coordinate

People do not decide alone. If one parent takes the only car, the other cannot, and someone must take the child to school. Does letting agents negotiate before leaving avoid these conflicts?

What this measures, and what it does not

It measures **coordination quality and cost**: how close each household's morning comes to the best possible plan, how many things go wrong, and how many LLM calls it takes. It does **not** measure whether the decisions are human-like. That requires real travel-survey data, as in the team's 2026 study, and is the natural next step.

HOW IT WORKS

A small world with real constraints

Five invented households: two parents with one car, an 8-year-old and a teenager; a family with two cars; a single parent whose 14-year-old can walk a younger brother to school; three flatmates sharing one car; a grandmother who is the only driver.

Everyone must reach work or school by a deadline. **Young children cannot travel alone. Cars are physical objects**: if one parent leaves at 07:00 with the car, the other finds no car at 07:45. Planned conflicts and real failures are counted separately.

MORNING EVENT	WHO KNOWS IT
Ordinary day	-
Bus strike · heavy rain	`journey planner`
Car warning light · ring-road accident · school or work closed · meeting moved earlier	`text message only`
Street party next Saturday	`trap: irrelevant`

Four ways of deciding

Rules

Habit + pick the cheapest, fastest option on the planner. Cannot read text.

LLM decides

A local LLM reads the situation and the news and chooses the option.

Gated hybrid

Rules by default; switches to the LLM when a text message arrives.

LLM reads, rules decide

The LLM only turns the message into a fact ("car unavailable"); rules decide with it.

Each one runs **without** negotiation (everyone plans alone) and **with** it. Two baselines without agents: a **central coordinator** that tries every family plan using the planner's information, and an **oracle** that also knows the hidden truth. Regret = distance from the oracle's plan, in minutes-equivalent. Every approach sees exactly the same information. All LLM outputs are validated, and fallbacks are labelled and counted.

Example: the bus strike in the Martin family

Without negotiation

Marie usually takes the bus, so she takes the car instead. Paul usually drives Léa to school.

07:00	`execute`	Marie leaves with the car.
07:45	`failure`	Paul: no car left at home (car at Purpan).

Paul walks Léa to school and arrives **95 min late** (work missed). Regret 537.

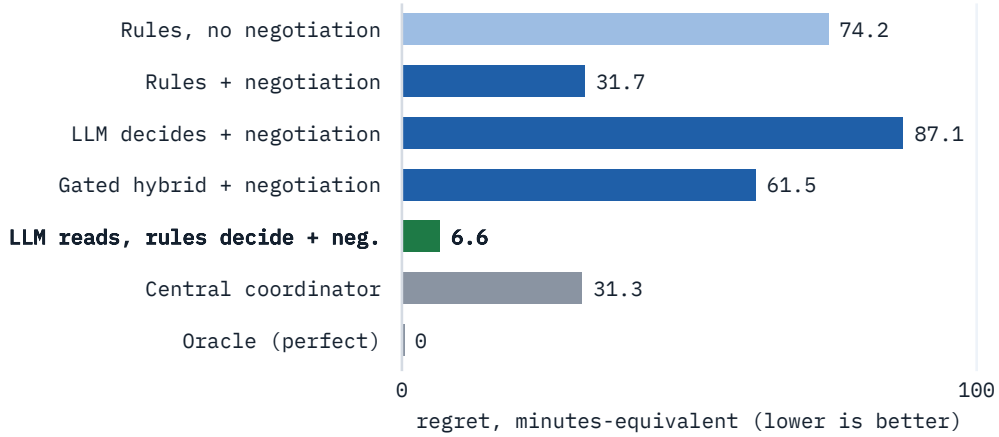
With negotiation (rule agents)

Léa	`request`	Who takes me to school by 08:20?
Marie	`propose`	Car, leaving 07:00.
Paul	`propose`	Car with Léa, leaving 07:45.
Marie	`counter`	Without the car I lose ~33.
Paul	`counter`	Without the car I lose ~545.
Marie	`accept`	I give up the car and cycle.
All	`commit`	Final plans.

Everyone arrives; Marie 5 min late. **Regret 0**, the best possible plan.

RESULTS · QWEN2.5:3B · 120 PAIRED SITUATIONS · 95% BOOTSTRAP INTERVALS

How close each approach came to the best plan



APPROACH (WITH NEGOTIATION)	REGRET (95% CI)	ACTIVITIES MISSED	THINGS THAT WENT WRONG / MORNING	LLM CALLS / MORNING
Rules	31.7 (20.0-45.5)	3.0%	0.25	0
LLM decides	87.1 (65.9-110.7)	14.6%	0.21	3.9
Gated hybrid	61.5 (44.6-79.6)	11.9%	0.21	2.4
LLM reads, rules decide	**6.6 (2.6-12.3)**	**1.4%**	**0.00**	**0.62**
Central coordinator (no agents)	31.3 (19.3-45.2)	3.5%	0.25	0

Five findings

1. **Negotiation fixes conflicts over shared resources, and simple rules are enough.** With it, rule agents halve their regret (74.2 to 31.7) and match the central coordinator on 118 of 120 situations.

2. **The small LLM is a poor decision-maker.** It does worse than rules on every kind of event, even when the news is text only. It often writes "the car is not reliable" and then plans to take the car. In negotiation it rarely gives way (9 deadlocks, settled by a logged tie-break rule).

3. **Calling the LLM only when news arrives saves calls, not quality.** The gated hybrid uses 38% fewer LLM calls but repeats the LLM's mistakes. The irrelevant trap news also triggers it and made 8 of 36 plans worse.

4. **Let the LLM read, and let the rules decide.** Translating the news into one fact per household, then deciding with rules and negotiation, gives the lowest regret (6.6). It beats the coordinator on 44 situations and loses on 2, and it never reacts to the trap news.

5. **It holds on new wording and a second model.** The news reader scored 74/80 on paraphrases it was never tuned on (qwen3:4b: 75/80). The larger qwen3:4b decides better than qwen2.5 (31.3 vs 56.6 on the same 40 situations) but is still worse than rules (19.8). The ranking of approaches is the same for both models.

EXAMPLE · THE CAR WARNING LIGHT (TEXT MESSAGE ONLY)

Why "LLM reads, rules decide" works

News: *"Last night the family car made a loud grinding noise and the engine warning light came on. The mechanic said on the phone not to drive it until it has been checked."* The journey planner still lists the car.

Rules

Cannot read the message. Paul plans to drive Léa. At 07:45 the car will not start. **Regret 140.**

LLM decides

Paul and Tom mention the warning. Marie still picks the car "as the fastest option" and loses 10 minutes when it will not start. Regret 10 here; on average across situations, the LLM decides worse than rules.

LLM reads, rules decide

One LLM call: evidence *"the car made a loud grinding noise"* → fact `car_unavailable`. Rules plan around it. **Regret 0.**

What it suggests

Separate **understanding** from **deciding**. The LLM is most useful when it turns open text into facts the simulation already handles: vehicle availability, cancelled activities, changed schedules. That is cheaper, easier to check, and keeps decisions consistent.

Communication and LLMs solve different problems. Negotiation removes **coordination** failures (who drives, who takes the child). Reading the news removes **information** failures (a broken car nobody told the planner about). The best result needs both: without negotiation, the best design's regret rises from 6.6 to 49.0.

Possible next steps in the team's platform

1. A news-interpreter step before the controller's feasibility filter.
2. The household negotiation step between itinerary listing and the probabilistic choice.
3. Evaluation against the EMC² survey, to test whether better coordination also means more realistic behaviour.

Limits, stated plainly

- Synthetic households, travel times and cost weights. Nothing here is calibrated on survey data or says how human the decisions are.
- The news reader picks from a fixed list of facts that matches what the simulator can produce: a best case.
- I wrote the scenarios and the held-out sentences myself. The trap news and the official events limit, but do not remove, that bias.
- Small local models at temperature 0; the second model was run on one third of the situations.
- Negotiation is typed (keep/give up, accept/decline), not free conversation.

Mistakes found and fixed

Inspecting the first run showed that the oracle could "optimally" leave a child at home, and that rule agents could not stay home when work was closed. Both were fixed and everything was re-run. The first run is kept in the repository with an explanation.

Code and full results: github.com/youssefdhouib/HybridMobility-Coop (Python, standard library only; one command reproduces every result) · Every number in this brief is generated from the result files by the scripts in the repository.

Builds on: Bru, Oberoi & Gaudou (2026), *Towards Adaptive and Real-Survey-Aligned Simulated Urban Mobility Behaviour of Generative Agents*, arXiv:2610.22407 (preprint submitted to AAMAS 2027) · Vu, Gaudou & Oberoi (2025), arXiv:2510.19497 · Taillandier et al. (2025), arXiv:2507.19364. Independent work, not affiliated with the laboratories; no code taken from their platform.